EXPERIMENT - 4

Hash Cracking Using John the Ripper

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux.
Login to the system.

Step 2: Open Terminal and Verify Tool

Open terminal and check if John the Ripper is installed:

john --version

Step 3: Create Hash File

Create a file and store a password hash:

nano hash.txt

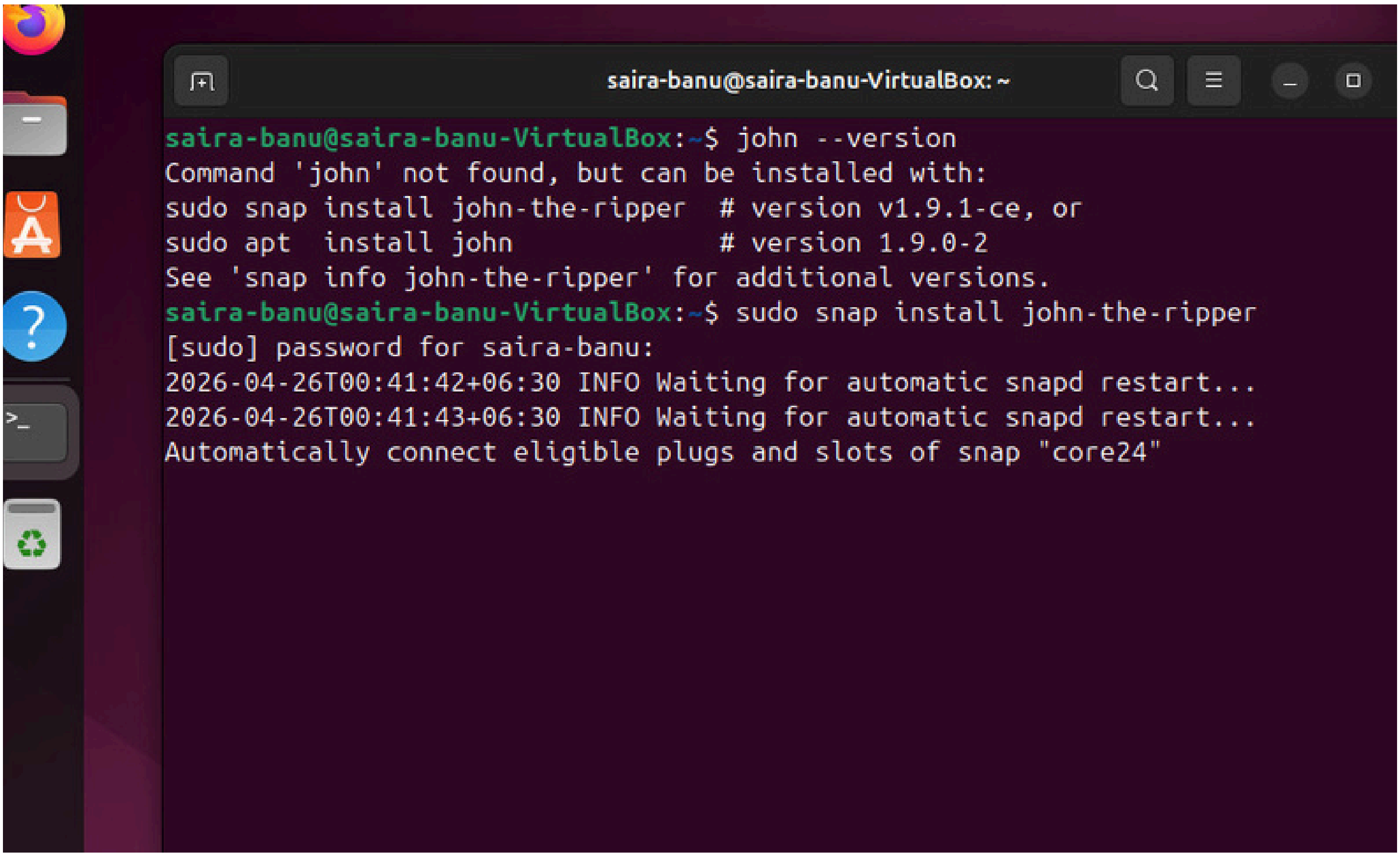
Step 4: Run Hash Cracking

john hash.txt

The tool attempts to crack the hash using default wordlists.

Step 5: View Cracked Password

john --show hash.txt


saira-banu@saira-banu-VirtualBox: ~
saira-banu@saira-banu-VirtualBox:~$ john --version
Command 'john' not found, but can be installed with:
sudo snap install john-the-ripper # version v1.9.1-ce, or
sudo apt install john   # version 1.9.0-2
See 'snap info john-the-ripper' for additional versions.
saira-banu@saira-banu-VirtualBox:~$ sudo snap install john-the-ripper
[sudo] password for saira-banu:
2026-04-26T00:41:42+06:30 INFO Waiting for automatic snapd restart...
2026-04-26T00:41:43+06:30 INFO Waiting for automatic snapd restart...
Automatically connect eligible plugs and slots of snap "core24"

```
Created directory: /home/saira-banu/snap/john-the-ripper/694/.john
Created directory: /home/saira-banu/snap/john-the-ripper/694/.john/opencl
Using default input encoding: UTF-8
No password hashes loaded (see FAQ)
saira-banu@saira-banu-VirtualBox:~$ nano hash.txt
saira-banu@saira-banu-VirtualBox:~$ john hash.txt
Using default input encoding: UTF-8
No password hashes loaded (see FAQ)
saira-banu@saira-banu-VirtualBox:~$ nano hash.txt
saira-banu@saira-banu-VirtualBox:~$ john --format=raw-md5 hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=4
Note: Passwords longer than 18 [worst case UTF-8] to 55 [ASCII] rejected
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/snap/john-the-ripper/current/bin/password.lst
Enabling duplicate candidate password suppressor
password         (?)
1g 0:00:00:02 DONE 2/3 (2026-04-26 00:53) 0.3891g/s 149.4p/s 149.4c/s 149.4C/s 123456..la
Use the "--show --format=Raw-MD5" options to display all of the cracked passwords reliabl
Session completed.
```

Result

Password hash cracking was successfully demonstrated, highlighting the need for strong passwords and secure hashing techniques.